

SLAM - Ex 4

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This document contains the answers to VAN Exercise 4. The code for this exercise is in [this link](#). The final version used for this submission corresponds to commit `cae40d8`.

In this exercise we extend the feature tracking across multiple frames and implement a suitable database for the tracked features. We use the matches we got in exercise 3 with careful bookkeeping to keep track (no pun intended) of which feature was tracked across what frame.

1 Tracking Database

We used the provided `tracking_database.py` implementation as the basis for the tracking database. The database assigns a unique `FrameId` to each stereo pair and a unique `TrackId` to each feature track. For each observation of a track in a frame, it stores a `Link` containing the stereo feature location as (x_l, x_r, y) , where (x_l, y) is the left-image location and (x_r, y) is the right-image location. Internally, the database keeps mappings from tracks to frames, from frames to tracks, and from each `(FrameId, TrackId)` pair to its corresponding `Link`. We used the implementation as provided, with a few small helper methods for querying track lengths and selecting tracks of sufficient length for the following sections.

2 Tracking Statistics

After constructing the tracking database, we computed several statistics over the resulting tracks. Trivial tracks of length one are not stored by our database implementation, since a track is created only after a feature is matched across two consecutive frames. Table 1 summarizes the statistics obtained for the full KITTI sequence.

Statistic	Value
Total number of tracks	518231
Number of frames	3300
Mean track length	3.32
Maximum track length	62
Minimum track length	2
Mean number of frame links	521.57

Table 1: Tracking statistics over the entire dataset (3300 frames).

3 Track Visualization

We selected a track of length at least 6 and visualized its feature locations in the left images. For each frame in the track, we cropped a 20×20 region around the feature, subject to image boundaries, and marked the feature location.

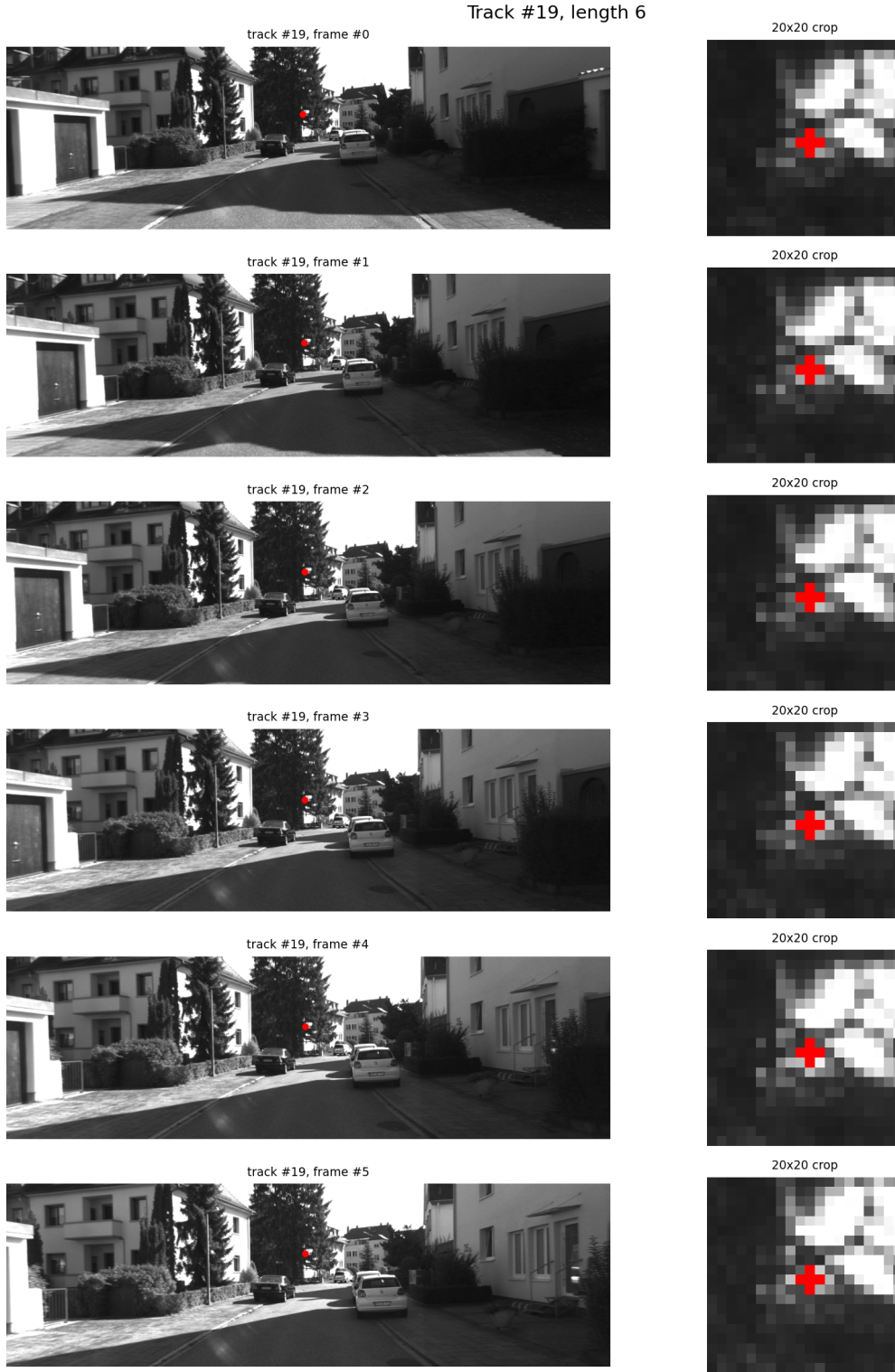


Figure 1: Visualization of a selected track of length at least 6. Each crop shows a 20×20 region around the tracked feature in a left image.

4 Connectivity Graph

For each frame i , we computed the number of outgoing tracks, defined as the number of tracks that appear both in frame i and in the consecutive frame $i + 1$:

$$|\text{Tracks}(i) \cap \text{Tracks}(i + 1)|.$$

This measures how many feature tracks successfully continue from one frame to the next.

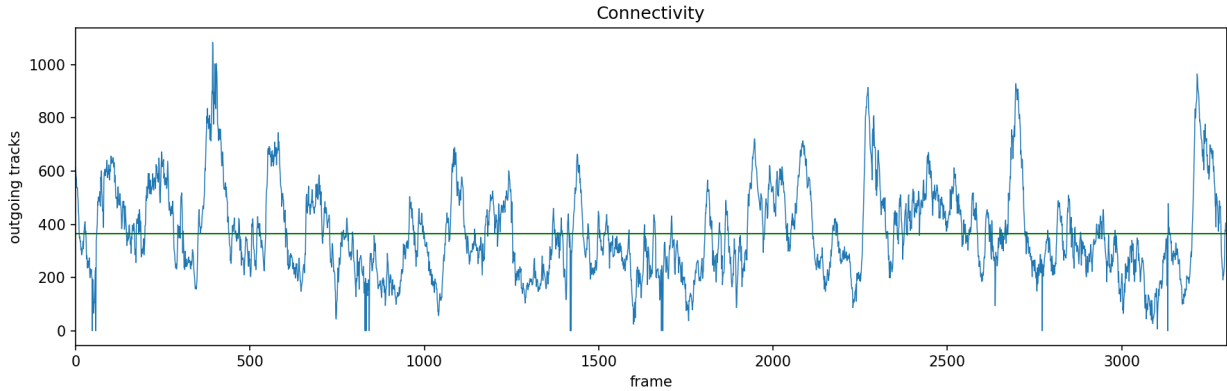


Figure 2: Connectivity graph showing the number of outgoing tracks for each frame.

5 Inlier Percentage

For each consecutive frame pair, we computed the percentage of temporal matches classified as inliers by the RANSAC-PnP stage. This measures the fraction of feature matches that are consistent with the estimated camera motion.

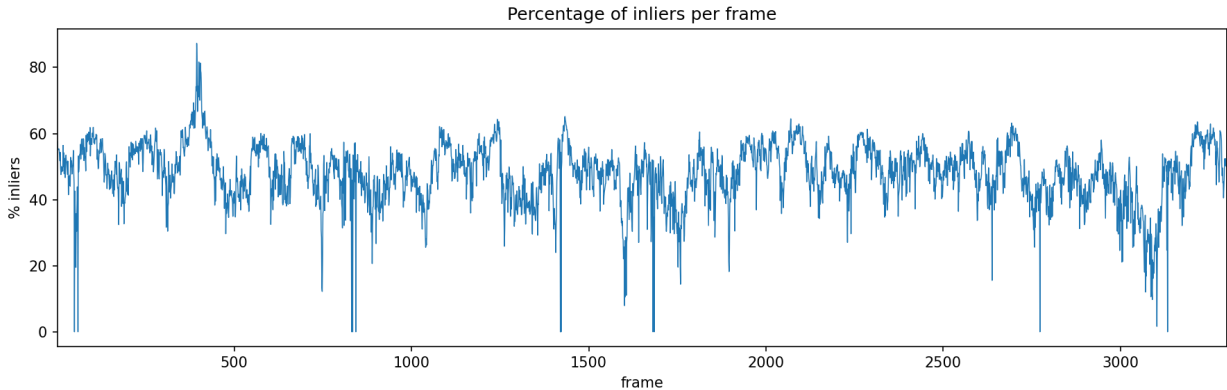


Figure 3: Percentage of RANSAC-PnP inliers for each consecutive frame pair.

6 Track Length Histogram

We computed the length of each track in the database and plotted a histogram of the resulting track lengths. The histogram shows how many tracks survived for each number of frames.

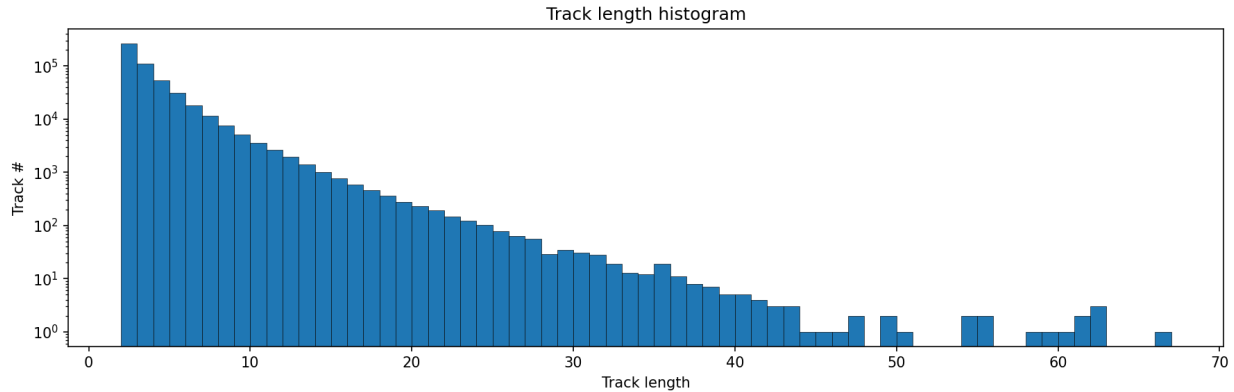


Figure 4: Histogram of track lengths in the tracking database.

7 Reprojection Error

We selected a random track of length at least 10 and triangulated its 3D point from the feature observations in the last frame, using the ground-truth camera poses. We then projected this point back to all frames in the track, for both the left and right cameras.

The reprojection error was computed as the L2 distance between the projected point and the tracked feature location. The results are shown in Figure 5.

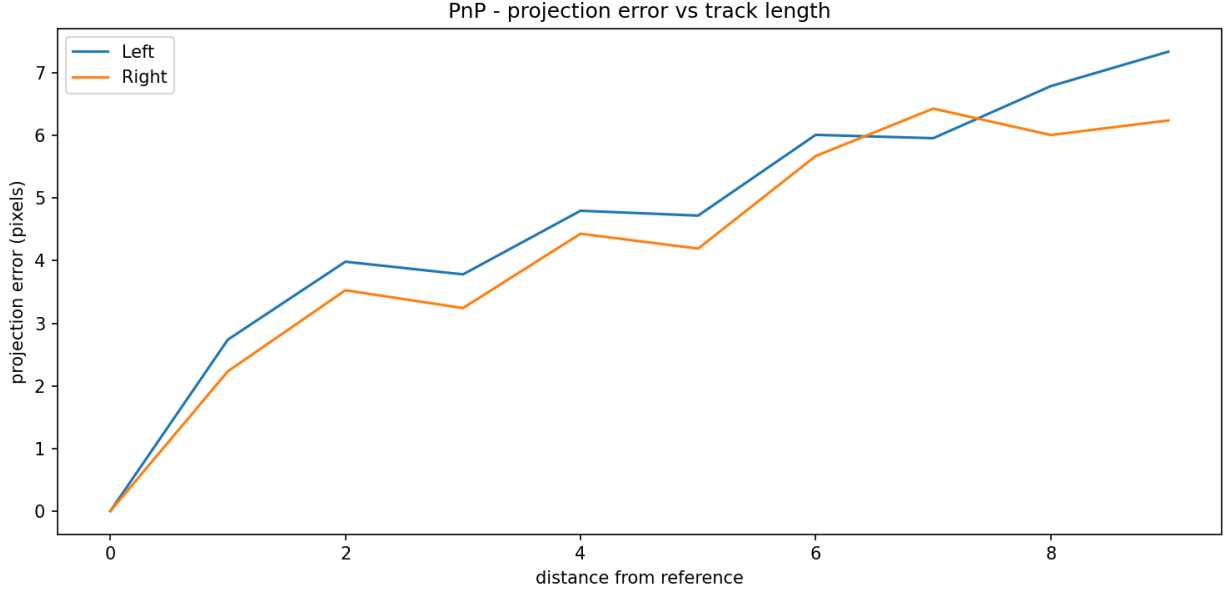


Figure 5: Reprojection error along a selected track. The 3D point was triangulated from the last frame and projected to the remaining frames using the ground-truth poses.

The error is smallest near the reference frame, since the 3D point was triangulated from that frame. As the distance from the reference frame increases, the error generally grows due to feature localization noise, triangulation uncertainty, and small pose or calibration errors. If we triangulated from the first frame instead, we would expect the error to be smallest near the first frame and increase farther along the track.

8 Random Gaussian Variables

Let $x \sim \mathcal{N}(\mu_x, \Sigma_x)$ and let $y = Ax + b$, where A is a nonsingular matrix. We compute the mean and covariance of y .

8.1 Mean and Covariance

Using the linearity of expectation, the mean of y is

$$\mu_y = \mathbb{E}[y] = \mathbb{E}[Ax + b] = A\mathbb{E}[x] + b = A\mu_x + b.$$

Next, by the definition of covariance,

$$\Sigma_y = \mathbb{E}[(y - \mu_y)(y - \mu_y)^T].$$

Substituting $y = Ax + b$ and $\mu_y = A\mu_x + b$, we get

$$y - \mu_y = (Ax + b) - (A\mu_x + b) = A(x - \mu_x).$$

Therefore,

$$\begin{aligned}\Sigma_y &= \mathbb{E} [A(x - \mu_x)(x - \mu_x)^T A^T] \\ &= A \mathbb{E} [(x - \mu_x)(x - \mu_x)^T] A^T \\ &= A \Sigma_x A^T.\end{aligned}$$

8.2 Show that y is Gaussian

It remains to show that y is normally distributed. Since A is nonsingular, the transformation $g(x) = Ax + b$ is invertible, and its inverse is $g^{-1}(y) = A^{-1}(y - b)$. The Jacobian of the inverse transformation is $J_{g^{-1}} = A^{-1}$,

and therefore

$$|J_{g^{-1}}| = |\det(A^{-1})| = \frac{1}{|\det(A)|}$$

The density of x is

$$f_x(x) = \frac{1}{\sqrt{(2\pi)^n |\Sigma_x|}} \exp \left(-\frac{1}{2} (x - \mu_x)^T \Sigma_x^{-1} (x - \mu_x) \right)$$

Using the random vector transformation formula,

$$f_y(y) = f_x(g^{-1}(y)) |J_{g^{-1}}|$$

Substituting $g^{-1}(y) = A^{-1}(y - b)$ gives

$$f_y(y) = \frac{1}{\sqrt{(2\pi)^n |\Sigma_x|}} \exp \left(-\frac{1}{2} (A^{-1}(y - b) - \mu_x)^T \Sigma_x^{-1} (A^{-1}(y - b) - \mu_x) \right) \frac{1}{|\det(A)|}.$$

Now note that

$$A^{-1}(y - b) - \mu_x = A^{-1}(y - b) - A^{-1}A\mu_x = A^{-1}(y - A\mu_x - b) = A^{-1}(y - \mu_y).$$

Therefore, the exponent becomes

$$\begin{aligned}& (A^{-1}(y - \mu_y))^T \Sigma_x^{-1} (A^{-1}(y - \mu_y)) \\ &= (y - \mu_y)^T A^{-T} \Sigma_x^{-1} A^{-1} (y - \mu_y).\end{aligned}$$

Since $\Sigma_y = A \Sigma_x A^T$, we have

$$\Sigma_y^{-1} = (A \Sigma_x A^T)^{-1} = A^{-T} \Sigma_x^{-1} A^{-1}.$$

Thus,

$$(A^{-1}(y - \mu_y))^T \Sigma_x^{-1} (A^{-1}(y - \mu_y)) = (y - \mu_y)^T \Sigma_y^{-1} (y - \mu_y).$$

It remains to rewrite the normalization constant. Since

$$|\Sigma_y| = |A\Sigma_x A^T| = |A||\Sigma_x||A^T| = |\det(A)|^2 |\Sigma_x|,$$

we get

$$\sqrt{|\Sigma_y|} = |\det(A)| \sqrt{|\Sigma_x|}.$$

Hence,

$$\frac{1}{\sqrt{(2\pi)^n |\Sigma_x|}} \frac{1}{|\det(A)|} = \frac{1}{\sqrt{(2\pi)^n |\Sigma_y|}}.$$

Combining the exponent and the normalization constant, we obtain

$$f_y(y) = \frac{1}{\sqrt{(2\pi)^n |\Sigma_y|}} \exp \left(-\frac{1}{2} (y - \mu_y)^T \Sigma_y^{-1} (y - \mu_y) \right).$$

This is exactly the density of a multivariate Gaussian random vector with mean μ_y and covariance Σ_y . Therefore,

$$y \sim \mathcal{N}(A\mu_x + b, A\Sigma_x A^T).$$